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Eliminating Cost and Space Barriers for Small-To-Midsize Emitters Through Fully Columnless and Modular Carbon Capture Technology

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Abstract

Scope: This paper presents Carbon Clean's CycloneCC™ technology, a fully modular and columnless post-combustion carbon capture solution designed to address the barriers facing heavy industry. By reducing capital expenditure and footprint, CycloneCC makes carbon capture more economically viable. The technology combines rotating packed beds (RPBs) and Carbon Clean's proprietary APBS-CDRMax® solvent. The goal is to demonstrate the use of first-of-a-kind (FOAK) technology in carbon capture and offer a cost-effective pathway to decarbonizing the hard-to-abate sector.

Methods: Carbon Clean has demonstrated the concept of modular intensified post-combustion carbon capture through pilot-scale and factory testing of RPBs and advanced solvent technology. CycloneCC combines two proven process intensification technologies:

- APBS-CDRMax, a proprietary amine-promoted buffer salt solvent.
- Rotating packed beds (RPBs), which utilise centrifugal force to enhance mass transfer.

The RPBs contain a disk of packing material that rotates around its axis. The centrifugal force generated through this rotational motion is significantly greater than the gravitational force seen in conventional columns. APBS-CDRMax has been developed by Carbon Clean and tested at the world's largest CO₂ capture demonstration facilities.

Conclusion: CycloneCC is currently the most developed modular carbon capture solution for industrial applications, enabling a 30% reduction in CapEx and OpEx. Compared to traditional solutions using 30% wt% MEA, APBS-CDRMax delivers a 20-fold improvement in corrosion resistance, 10-fold improvement in solvent degradation, and 30% lower energy consumption. The process intensification of the technology has reduced footprint by 50%, height by 70% and steel usage by 35%.

An industrial-scale demonstration with a leading energy company was conducted at the Al Ruawais Industrial Complex in Abu Dhabi. The unit was installed within a week due to the prefabricated design, which reduced operational disruption and safety risks. The unit operated for around 4,000 operating hours over a six-month period, capturing CO₂ from a reformer flue gas stack and utilizing it in urea production.

CycloneCC is now at Technology Readiness Level (TRL) 7 and has been deployed at a site in Saudi Arabia and Canada with key partners for further validation in varied climates and at different CO₂ concentrations in the inlet flue gas.

The use of AI has allowed reliable adjustments, with human operators implementing AI-suggested recommendations to increase reliability and availability.

Recent factory testing confirmed CycloneCC C1 RPBs' mechanical performance, including the ability to capture 285 tonnes of CO₂ per day, representing a 20-fold RPB scale-up from the first industrial demonstration of CycloneCC.

Additive Information: Carbon Clean will provide FEED support to maritime decarbonization with an engineering partner. The study envisages the installation of a CycloneCC unit on an FPSO vessel as a pilot, and it will be a first-of-a-kind deployment for onboard carbon capture.

CycloneCC is well-suited to an offshore environment, as the unit delivers a height reduction of 70% when compared to conventional solutions. The RPBs will achieve enhanced capture performance under vessel motions, making CycloneCC ideal for offshore operations.

Introduction

This paper presents performance data and operational insights from the industrial-scale deployment of Carbon Clean's CycloneCC unit at a nitrogen fertiliser facility in Abu Dhabi. The demonstration builds upon prior pilot-scale evaluations and published findings ([Bumb et al., 2024](#)), showcasing advancements in modular carbon capture design and process intensification.

CycloneCC is a fully modular, skid-mounted post-combustion carbon capture system engineered to address two principal barriers to carbon capture adoption in industrial settings: capital expenditure and spatial constraints. The system integrates rotating packed bed (RPB) absorbers with Carbon Clean's proprietary amine-promoted buffer salt solvent (APBS-CDRMax), enabling a tenfold reduction in equipment size and up to a 50% footprint minimization compared to conventional MEA-based systems.

Carbon Clean has established itself as a technology leader in the carbon capture, utilisation, and storage (CCUS) sector, particularly within hard-to-abate industries such as cement, steel, refining, and energy-from-waste. With over a decade of operational experience, the company has delivered more than 50 technology references globally and maintains a robust intellectual property portfolio comprising 110 active patent assets across 18 families in 30 countries.

The organisation's growth is supported by public and private sector partnerships, including funding from UK and US government initiatives and collaborations with heavy industry partners. Carbon Clean's Global Innovation Centre (GIC), located in Navi Mumbai, India, serves as a dedicated research and development hub for CCUS technologies. The facility spans over 77,000 square feet and houses state-of-the-art solvent testing laboratories, pilot-scale capture plants, and a multidisciplinary engineering team with expansion capacity to 350 personnel.

The GIC plays a pivotal role in scaling CycloneCC for commercial deployment, accelerating technology readiness through iterative design validation and solvent optimization. The modular architecture of CycloneCC enables rapid installation and flexible integration across diverse industrial sites, offering a viable decarbonization pathway aligned with global climate objectives.

Statement of Theory and Definitions

Carbon Clean has successfully demonstrated the concept of modular intensified post-combustion carbon capture, defined as the removal of CO₂ from flue gases following the combustion of a fuel source ([Chao et al., 2021](#)). This was achieved through laboratory-scale and pilot-scale testing of Rotating Packed Beds

(RPBs) and proprietary solvent systems. The company has since commissioned the first industrial-scale CycloneCC demonstration unit, representing a tenfold scale-up from its 1-ton-per-day (TPD) pilot system.

Decarbonization refers to the reduction or elimination of carbon dioxide emissions, a critical objective in achieving global net-zero targets. CycloneCC introduces a paradigm shift in post-combustion capture by integrating two advanced process intensification technologies: a proprietary amine-promoted buffer salt solvent (APBS-CDRMax) and RPBs, which replace conventional static absorption columns to significantly reduce equipment footprint.

RPBs are process intensification devices that utilize centrifugal force to enhance gas-liquid mass transfer. Operating under high-gravity conditions, they increase interfacial area and reduce equipment size, thereby improving CO₂ absorption efficiency (Burns and Ramshaw, 1996). This intensified approach enables compact, high-performance capture systems suitable for small to mid-scale industrial emitters.

Modular technology is defined as a design methodology that divides systems into interchangeable, self-contained units (modules). These modules can be independently developed, replaced, or upgraded without affecting the overall system architecture. Modular design theory, introduced by Starr (1965), and product architecture theory (Ulrich and Eppinger, 1995) support the scalability and flexibility of CycloneCC across diverse industrial sectors, including cement, steel, energy, and oil and gas.

Traditional post-combustion capture systems rely on large absorber and desorber towers using monoethanolamine (MEA) at 30 wt.%, a benchmark solvent with known limitations in energy intensity, cost, and spatial requirements (Luis, 2016). In contrast, CycloneCC's modular and intensified configuration offers a compact, scalable alternative with reduced operational complexity.

Innovation, in this context, is defined as the creation and implementation of new technologies or processes that deliver significant improvements in performance, efficiency, or sustainability. According to Schumpeter's theory of creative destruction (1934), such innovations disrupt existing paradigms and enable transformative industrial practices. CycloneCC exemplifies this by combining novel solvent chemistry with intensified reactor design to advance carbon capture capabilities.

CycloneCC: the process

RPBs have been utilized for commercial purposes since the 1960s. However, post-combustion CO₂ capture is a new use for RPBs. The RPB is made up of a rotor that contains packing material and is powered by a motor linked to the rotor shaft. The rotor and packing material are enclosed in a casing with an inlet and exit for both liquid and gas streams.

CycloneCC uses RPBs to perform crucial column operations in a typical solvent capture process. The coupling of RPBs with the unique commercial APBS-CDRMax solvent in the absorber and desorber unit operations improves carbon capture efficiency.

A render of the RPB configuration is shown in Fig. 1 and the CycloneCC process flow diagram for the latest scaled-up commercialized design Fig. 2.



Figure 1—RPB render

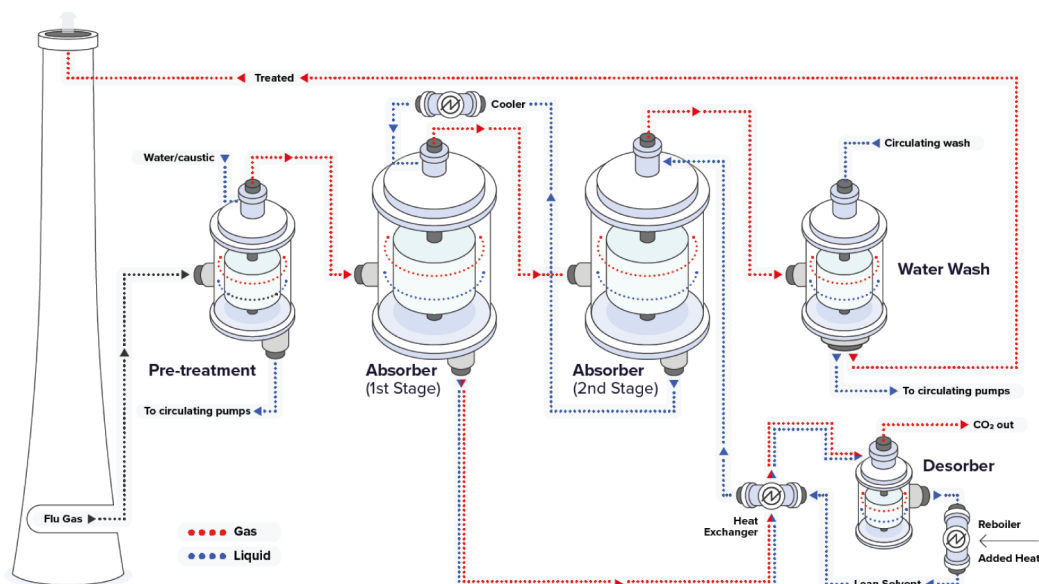


Figure 2—CycloneCC process flow diagram

Rotating Pack Beds (RPBs)

RPBs by rotating on an axis produce a centrifugal force, which is reliant on mass, angular velocity, and radius, and radiates outward from the centre of the packed bed. Upon coming into contact with the packing, the APBS-CDRMax solvent is driven radially towards the packing's outer edge by the centrifugal force of the RPB. After being gathered in the outer house, the solvent drains to a sump and is then pumped to the following step in the procedure.

- RPB absorber: The solvent enters the absorber through the inner radius after the CO₂ rich flue gas stream is injected from the outer radius and passes through the packing material. As a result, there is a counter-current interaction between the gas and the liquid. The solvent specifically absorbs CO₂. A liquid distributor at the inner radius feeds the RPB absorber with Carbon Clean's APBS-CDRMax solvent.
- RPB desorber: The solvent containing CO₂ enters the RPB desorber and comes into indirect contact with steam. The CO₂-depleted APBS-CDRMax solvent then returns to the RPB absorber to resume the capture process once CO₂ has been removed from the solvent.

Process intensification

According to Moulijn and Syankiewicz (2000), Carbon Clean has shown that chemical engineering may result in a significantly smaller, cleaner, and more energy-efficient technology. RPBs perform mass transfer operations far more effectively than traditional packed bed columns because of the centrifugal force produced by the packed bed's rotating motion, which is far higher.

The mass transfer improvement allows the RPB to be up to 10 times smaller than traditional columns to accomplish the same results. The combination of RPBs and APBS-CDRMax provides better mass and heat transfer between the liquid and gas phases through thinner liquid films produced by a centrifugal force. In addition, a more intense turbulent flow is relative to conventional columns. The liquid films and droplets created in the packing material are remarkably thinner, which increases the surface area to volume ratio of the liquid. This results in faster and higher mass transfer efficiency between the gas and liquid phases.

Summary of the CycloneCC development, and the next steps required towards commercialization.

- 2015: Starting bench-scale RPB work and building RPB absorber prototype at a nominal size

- 2016: Testing of bench-scale RPB and confirming RPB synergy with APBS-CDRMax solvent and benchmarking vs. MEA
- 2018: Design and development of 1 TPD absorber and desorber, creating RPB scale-up rules, and designing RPB for >90% CO₂ capture
- 2020: First testing of 1 TPD RPB absorber and desorber, validating RPB scale-up rules, parametric testing on inlet conditions, and packing testing
- 2021: Scaling absorber and desorber for mobile system and recruiting a technology validation partner for testing
- 2023: Development and testing at industrial demonstration scale with industrial flue gas
- 2024: CycloneCC market entry at TRL 7, developing to 9
- 2025: CycloneCC C1 factory testing completion

Description and Application of Equipment and Processes

The modular design of CycloneCC RPBs is made possible by their reduced packing volume, which is 10 times lower than that of a static absorber column with an equivalent capacity. The unit is supplied in twelve road transportable sections after being prefabricated off-site (see [figure 3](#)). As a result, less infrastructure is needed on the job site, and installation takes less time. With its smaller size and up to 50% lower installation costs, CycloneCC is the perfect carbon capture option for brownfield sites or locations with restricted space.

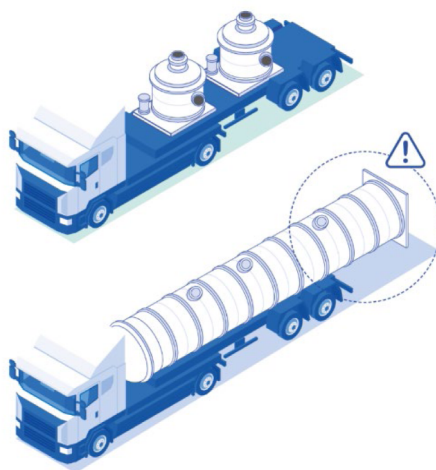


Figure 3—RPB vs conventional absorber transportation

RPB laboratory-scale testing (TRL 3-4)

RPB development began by building a laboratory-scale RPB absorber prototype. The first laboratory-scale prototype was constructed to demonstrate the feasibility of applying RPB technology to CO₂ capture and to benchmark various solvents. See below for project details and setup images.

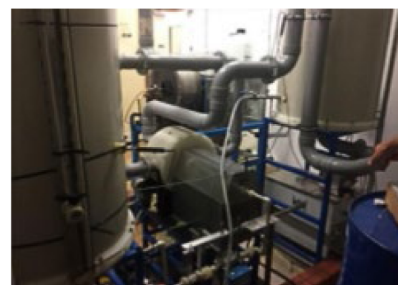
Project overview

- Goals: Test RPB and APBS-CDRMax solvent performance and demonstrate the order of magnitude size reduction of RPB vs. a static conventional column.
- Timeline: 2015 - 2017.
- Success parameters: Use of RPB absorber in CO₂ capture and establish design rules for RPB sizing.

- Key R&D endeavours: Build, test, and validate an RPB prototype and benchmark RPB CO₂ capture with APBS-CDRMax vs. MEA.



(a)



(b)

Figure 4—Lab scale RPB setup (a) front view of RPB, (b) back view of RPB. (Kolawole, 2021)

Collectively, the learnings gained about the process and mechanical conditions, along with the confirmation of RPBs' size advantage over conventional columns, warranted a scale-up of the technology to 1 TPD to define and validate scale-up rules used to size the RPB for 90% CO₂ capture.

RPB pilot-scale testing (TRL 5-6)

Carbon Clean developed a beta prototype project at a scale of 1 TPD of CO₂ in the UK. With the support of a grant from the UK Government's Department for Business, Energy & Industrial Strategy (BEIS), Carbon Clean built and demonstrated a 1 TPD pilot plant to capture CO₂ from modulated flue gas using CycloneCC technology.

The pilot plant comprised an RPB absorber, a novel RPB regeneration unit, and an integrated stripper reboiler (ISR), which had been sized to achieve 90% CO₂ capture. These pilot plant units are integrated into a conventional packed-bed pilot plant and can be operated independently of one another, allowing for the optimization of each unit. See below for the project details (Frost, 2024).

Project overview

- Goals: Demonstrate a 90% CO₂ capture rate on flue gases with varying CO₂ content, to prove the order of magnitude size reduction of RPB absorber and desorber, and show reduction in CapEx and OpEx for CO₂ capture.
- Timeline: April 2018 - March 2022.
- Key IP: Demonstration of the integrated stripper reboiler (ISR), which allows for improved heat and mass transfer in the RPB desorber.
- Success parameters: Integrated use of RPB absorber and desorber in a single system, validation of scaling rules for RPB sizing, achieving 90% CO₂ capture efficiency, and achieving solvent regeneration using an integrated desorber reboiler.
- Key R&D endeavors: Perform parametric testing over a range of liquid inlet conditions (flow rate, composition, temperature) to improve understanding of the impact on CO₂ capture. Test a range of solvents at varied concentrations to understand their efficiency in capturing CO₂ at a pilot plant and explore different types of packing to understand their individual impact on CO₂ capture at a pilot plant.

APBS-CDRMax solvent

With amine-promoted buffer salts (APBS), a proprietary mixture of amines and salts, Carbon Clean has enhanced traditional solvents. This sophisticated reaction chemistry provides the low regeneration energy of

a buffer salt with the strong kinetic reactivity of an amine in a unique way. The end product is a special high-capacity, fast-acting carbon capture solvent. APBS-CDRMax, Carbon Clean's patented CO₂ capture solvent, was developed to maximize carbon capture efficiency. By selectively absorbing CO₂ into the reactive solvent components, the solvent chemistry enables the quick extraction of carbon dioxide from flue gases with CO₂ concentrations ranging from 3 to 20% vol.%.

The active amine that absorbs CO₂ through a promoter-based mechanism is released by the buffer salt system, which has a large capacity for CO₂ and lowers the energy needed for the CO₂ capture process. Because of this, APBS-CDRMax generates CO₂ with a dry basis purity of ≥ 99.5 vol%, which lowers corrosivity, increases stability, and requires less regeneration energy. Carbon collection and utilization initiatives in the US, UK, Netherlands, Sweden, Norway, Spain, and India have made use of APBS-CDRMax. Based on the initial vapor-liquid equilibrium measurements, kinetics, and solvent management considerations, specifically, solvent degradation, corrosion, emissions, and continuous absorption and desorption testing. Carbon Clean conducted a thorough research and development program for solvent characterization and optimization.

APBS-CDRMax benefits

- Higher delivery of CO₂ loading factor than industrial counterparts, providing a reduction in solvent circulation rate, smaller equipment, and less energy consumption for regeneration.
- 20x less corrosion, 10x less degradation, and less reaction to oxygen when compared to conventional solvents, helping lower waste solvent disposal costs to a negligible amount.
- 5x longer solvent life and 86% less solvent makeup.
- A higher performance efficiency with less foaming, leading to a 50% reduction in ongoing chemical requirement and waste disposal costs, reducing amine carryover and the need for anti-foaming additives.
- A reduction in solvent emissions to parts per billion (ppb) levels, which meets environmental regulatory requirements and facilitates approval.
- Tested and proven results with flue gases from natural gas and solid fuel combustion, waste-to-energy, steel production, cement production, and refineries.

CycloneCC scale up (TRL 7)

The first deployment of an industrial demonstration CycloneCC unit was in a nitrogen fertilizer facility in Abu Dhabi, United Arab Emirates, where CycloneCC was chosen by a leading energy company for a carbon capture pilot. Enhancing CycloneCC scale-up correlations and testing the technology in an industrial setting were the goals of this research. A 10 vol.% CO₂ content in the gas to be treated is the target for this integrated transportable CycloneCC pilot plant. For convenience of assembly and transportation, all process equipment is mounted on road truckable steel frames. The project's transportable unit will enable CycloneCC technology to be validated in actual industrial settings at several host sites. This will provide operational understanding, data to establish detailed techno-economic evaluations for various industrial flue gas sources and reduce risk in future commercial deployment.

Project overview

- Goals: Demonstrate the technology in real industrial settings within heavy industry, approximately 50% cost reduction concerning the benchmark technology, including both CapEx and OpEx, and a significant miniaturization of the CO₂ capture process equipment - reducing the plant footprint by 50% compared to existing conventional technology.
- Timeline: October 2023 - 2024.

- Key IP: Integrated CO₂ capture process.

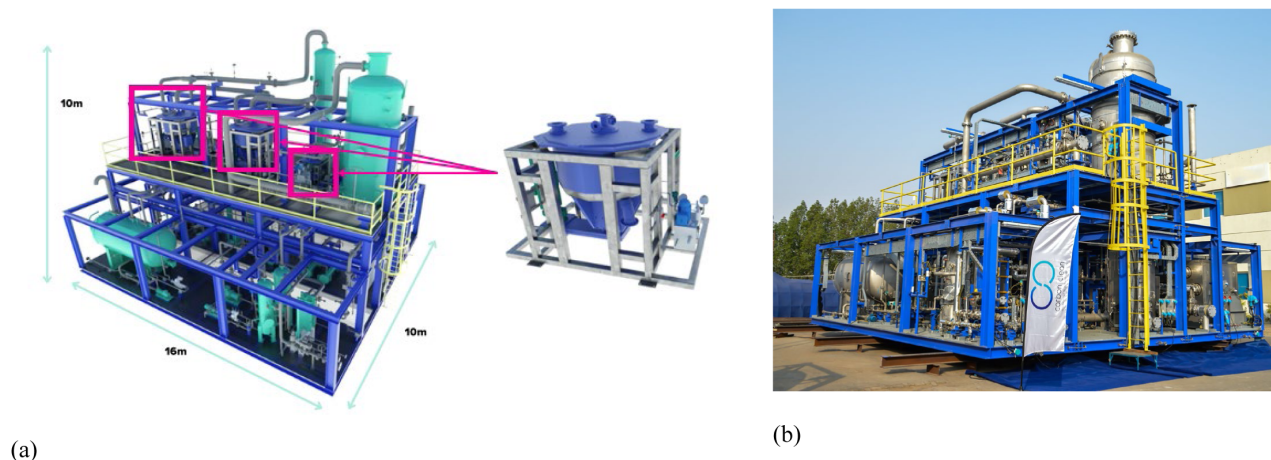


Figure 5—(a) CycloneCC industrial demonstration unit design and (b) CycloneCC industrial demonstration unit.

The CycloneCC industrial unit is designed to function optimally when processing flue gas with CO₂ concentration in the range of 3-20 mol%.

The modules were delivered to the site with fully installed piping, instrumentation, cabling, and equipment, having first undergone factory testing and a trial assembly - the standardized, prefabricated unit was installed on site in under a week, ready to be commissioned.

Demonstration challenges

The demonstration project with the partner lasted 6 months, during which approximately 1,000 tonnes of captured CO₂ were fed into a Urea Plant for utilization at >99.7% purity. The trial began in June and had challenges reaching the required CO₂ production rate of 7 TPD. To overcome this, troubleshooting, root cause analysis and optimization activities were performed to increase production to 7TPD. After performing all the recommended activities. On December 1 2024, the demonstration unit reached the required 7TPD with 90% CO₂ recovery within a week.

Summary of challenges

- Extreme weather conditions (55-65°C)
- Simplified/new design approach to aid compactness and reflect demonstration rather than commercial plant
- Instrumentation selection and positioning, harder for troubleshooting
- Faster dynamics on level maintenance in RPBs
- Instrumentation failures (pumps and instruments) due to extreme weather conditions (this does not include the RPBs, which performed successfully throughout the demonstration), as the unit was originally designed for European conditions
- Compact Reboiler design
- High Cooling water supply temperature (45-48°C) than the designed value
- High Steam supply temperature (260°C) than required at the BLs (175°C).
- Compact water wash design to aid emissions reduction

Innovative use of Carbon Clean-developed Artificial Intelligence (AI) has contributed to the plant's increased reliability and availability, as well as maximizing prediction production and overcoming many challenges.

The integration of AI into the CO₂ capture system has significantly reduced operational costs through continuous process optimization and enhanced equipment uptime.

This is achieved via real-time monitoring and predictive diagnostics that identify potential failures before they occur. Every 30 minutes, the AI system generates actionable recommendations that are delivered to workstation operators, enabling them to implement targeted adjustments that improve CO₂ production and efficiency.

Critical plant signals, such as vibration patterns, temperature fluctuations, and pressure readings associated with the RPB's stability, are collected at one-minute intervals. These inputs are analyzed by the AI engine to generate anomaly scores and assess overall plant health, allowing early detection of irregularities and flagging of events that warrant further investigation.

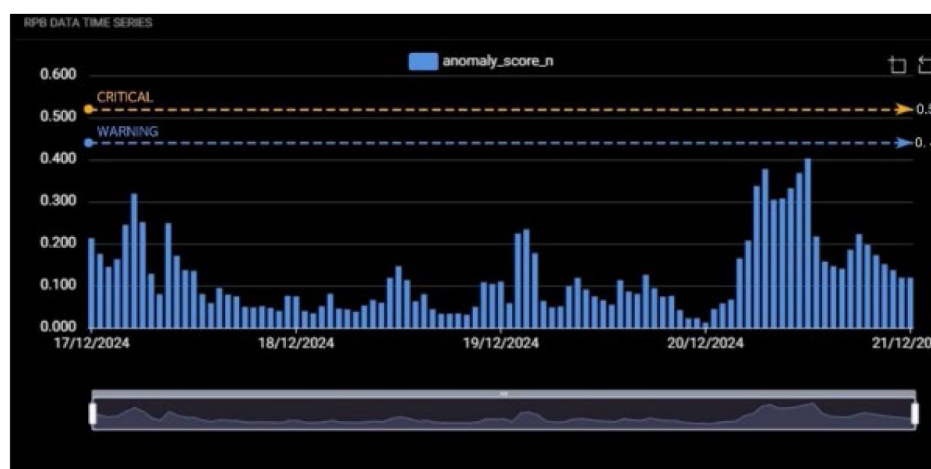


Figure 6—Example RPB anomaly dashboard

Due to the success of the demonstration unit in 2024, a leading energy company in Saudi Arabia signed a collaboration agreement to demonstrate CycloneCC technology. This demonstration will further validate Carbon Clean's novel CycloneCC technology to capture CO₂ from natural gas turbine exhaust streams containing approximately 4% CO₂. This first-of-a-kind deployment, capturing very low concentrations of CO₂, is a key milestone in scaling up and commercializing CycloneCC.

CycloneCC C1 (TRL 8)

Recent developments in rotating packed bed (RPB) design have culminated in the successful factory-scale testing of one of the largest RPB units built to date for post-combustion carbon capture. The system utilizes centrifugal force to intensify gas-liquid contact, thereby enhancing CO₂ absorption efficiency. Under design conditions, the unit is capable of capturing up to 285 tonnes of CO₂ per day.

This advancement reflects a more than 20-fold scale-up relative to earlier industrial demonstrations, which targeted capture rates of approximately 10 tonnes per day. The current RPB configuration integrates with an amine-promoted buffer salt solvent system, offering increased absorption kinetics and reduced regeneration duty compared to conventional benchmarks.

The CycloneCC C1 series replaces standard packed column absorbers with compact RPB-based modules, resulting in significant reductions in physical dimensions and material requirements. Each C1 unit is designed to capture up to 100,000 tonnes of CO₂ per annum. Compared to traditional systems, equipment height is reduced by up to 70%, steel usage by approximately 35%, and overall footprint by nearly 50%.

Major components exhibit a volumetric reduction factor of up to ten when compared with conventional absorber and desorber equipment.

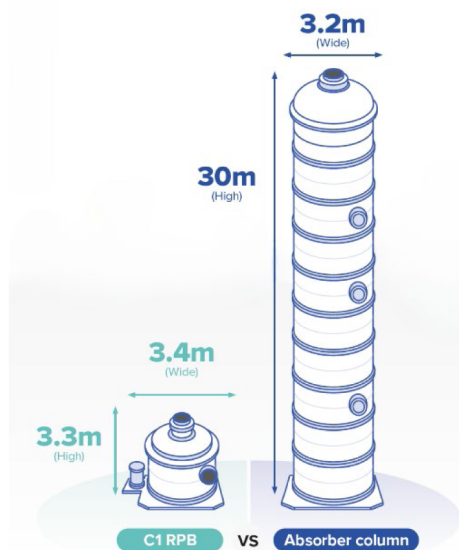


Figure 7—C1 RPB comparison to conventional absorber columns

This configuration offers a viable alternative for facilities seeking intensified capture performance within constrained spaces, with the potential to facilitate modular integration and rapid deployment across diverse industrial applications.

Maritime operations

Ongoing collaborations aim to establish a structured framework for adapting intensified carbon capture processes to offshore maritime operations. Carbon Clean is engaged with key stakeholders to assess and develop integration pathways for the CycloneCC system on Floating Production Storage and Offloading (FPSO) platforms.

The phased approach comprises an initial pilot installation scheduled for deployment on an FPSO by 2026. This pilot will support the evaluation of system performance under offshore operational constraints. Subsequent scale-up targets include partial decarbonization via CO₂ capture rates of up to 100,000 tonnes per annum, followed by integration into full FPSO design architecture with system capacity approaching 300,000 tonnes of CO₂ captured annually.

Longer-term objectives involve the deployment of carbon capture systems capable of processing up to 1,000 tonnes of CO₂ per day onboard FPSO vessels. These efforts seek to define technically feasible solutions for maritime applications where space limitations, dynamic vessel conditions, and process stability present significant challenges to land-based system transferability.

This development pathway follows preliminary engineering activities undertaken in collaboration with our partner, including a Front- End Engineering and Design (FEED) study initiated in early 2025. That effort supports the first installation of a CycloneCC unit on an FPSO, enabling the systematic evaluation of offshore carbon capture system integration.

CycloneCC's suitability for maritime deployment is based on its compact design architecture. Compared to conventional column-based configurations, RPB-based units demonstrate a footprint reduction of approximately 50%, with absorber and desorber heights reduced by up to 70% in the latest design iteration (C1 series). The rotating packed bed design offers enhanced hydrodynamic and mass transfer stability under vessel motion, key considerations for FPSO operations.

Global Innovation Centre

Launched in July 2025, Carbon Clean's Global Innovation Centre (GIC), located in Navi Mumbai, India, is positioned to become a major international hub for the development of next-generation carbon capture technologies. The facility spans 77,121 square feet and is designed to accommodate up to 350 personnel. It supports the full innovation lifecycle, from initial concept (TRL 0) to commercial deployment (TRL 7).

The GIC plays a critical role in accelerating technological advancement by integrating laboratory-scale research, prototype validation, data generation for model development, and scale-up activities. It serves as a comprehensive platform for transforming early-stage ideas into commercially viable solutions.

Key Facilities and Capabilities

- **Process and Solvent Development Laboratory:** Dedicated to the formulation and optimization of advanced solvents, including proprietary NAS solvent, and the development of emissions control technologies.
- **Analytical Laboratory:** Provides support for after-sales services and conducts detailed solvent analysis to ensure performance reliability and regulatory compliance.
- **Pilot Plant Infrastructure:** Enables process design validation for solvents and facilitates scale-up testing of rotating packed bed (RPB) technologies (see [figure 8](#)).
- **Utilities and Support Systems:** Comprehensive auxiliary systems and utilities to support pilot plant operations and ensure seamless integration of experimental workflows.
- **Solvent Development Laboratory for VLE Mapping:** Equipped to generate vapor-liquid equilibrium (VLE) data, which is essential for accurate process modeling and solvent performance evaluation.



Figure 8—Mico pilot carbon capture plant within the GIC

Presentation of Data and Results

In December 2024, the CycloneCC industrial demonstration carbon capture system achieved a milestone of 4,000 cumulative operating hours, reaching Technology Readiness Level 7 (TRL 7). The system was installed at a nitrogen fertilizer production facility in the Al Ruwais Industrial Complex, Abu Dhabi, UAE. The CO₂ is captured from the reformer flue gas stream and utilized in downstream urea synthesis.

The CycloneCC unit integrates a proprietary amine-promoted buffer salt solvent (APBS-CDRMax) with Rotating Packed Bed (RPB) technology, replacing conventional absorber and stripper columns. This process intensification approach enables a reduction in equipment volume by a factor of 10.

System validation confirms that the captured CO₂ meets the required purity specifications for urea synthesis. The RPBs demonstrated stable hydrodynamic performance and effective gas-liquid mass transfer under industrial flow conditions. Solvent degradation was minimal, and corrosion rates were significantly

lower than those observed in monoethanolamine-based systems. The system confirmed that CO₂ recovery, energy consumption, and CO₂ purity were simultaneously achieved.

The modular, skid-mounted configuration of CycloneCC allows for prefabrication and factory acceptance testing prior to site delivery. This design minimizes civil works and enables rapid deployment. The columnless architecture reduces structural height by 70% and steel usage by 35%, further contributing to space and material efficiency.

Operational data supports continuous performance, with plans to optimize throughput and validate extended runtime. These results confirm the system's scalability and reliability for industrial applications. The results have supported the next project at a Saudi Arabian site, working with natural gas turbines, to validate the technology at low concentration. Additionally, the results support Carbon Clean's acceleration of the C1 series, which can capture up to 100,000 tonnes of CO₂ annually.

Carbon Clean will continue to develop the AI algorithms by implementing strategies to improve the rate of learning, and will also focus on developing transfer learning, which will increase the speed of deployment of the platform across multiple sites.

Factory Acceptance Testing

In May 2025, CycloneCC C1 RPB passed factory testing, verifying mechanical performance, including that the technology can capture up to 285 tonnes of CO₂ per day. It represents a more than 20-fold RPB scale-up from the first industrial demonstration of CycloneCC, which was designed to capture up to 10 tonnes of CO₂ per day.



Figure 9—RPB at Thomas Broadbent, UK

Conclusions

This study presents an intensified approach to carbon capture based on the integration of rotating packed beds (RPBs) with an amine- promoted buffer salt solvent (APBS-CDRMax). The configuration supports process intensification by enabling significant reductions in equipment scale and energy demand while maintaining high CO₂ capture efficiency.

The technology adopts a modular, column-free design that facilitates pre-fabrication and rapid on-site installation, offering a deployment pathway for post-combustion capture applications in space-constrained facilities. It is engineered to address direct emissions (Scope 1) from industrial sectors, including oil and gas, cement, steel, and chemical processing, which represent a substantial share of global emissions.

Performance validation was achieved through a long-duration field demonstration in the United Arab Emirates, during which the system operated for 4,000 hours with sustained capture efficiencies up to 90%. Captured CO₂ was applied in downstream synthesis operations, reinforcing the technical feasibility of circular carbon utilization.

Capital cost reductions are attributed to the use of RPBs and high-performance packing materials. Lower solvent inventory and smaller auxiliary equipment further contribute to reduced installed costs. Modularity and standardization offer additional opportunities for economies of scale in manufacturing and deployment.

Operational expenditure savings result primarily from the reduced regeneration duty of the APBS-CDRMax solvent compared to monoethanolamine (MEA), along with enhanced solvent stability, reduced corrosion propensity, and lower solvent loss rates. Further efficiency gains are observed in reduced pumping and cooling requirements, which contribute to the overall energy performance of the process.

This work outlines the development trajectory from laboratory-scale optimization to industrial-scale validation, establishing a platform for intensified carbon capture in industrial settings.

Acknowledgements

The authors acknowledge the contributions of the Carbon Clean technical team involved in the design, testing, and deployment of the carbon capture system described in this study.

Support provided by the U.K. Department for Business, Energy and Industrial Strategy is gratefully acknowledged. Technical collaboration with Newcastle University, the University of Sheffield, Doosan Babcock Ltd., and Broadbent Ltd. has played an important role in advancing the evaluation and testing of the system.

The authors further recognize the commitment of participating stakeholders in enabling deployment-scale testing and system integration required for accelerated technology readiness.

This paper represents a follow-up on ‘CycloneCC: The Technology to Accelerate Industrial Decarbonisation’ (Bumb et al. 2024), which was presented at ADIPEC 2024. Please note that the original authors of that paper are included in this paper.

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Appendix



Figure 1—RPB render

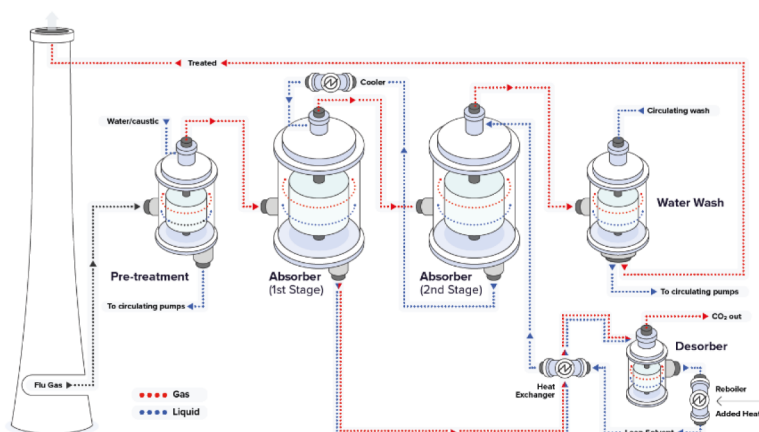


Figure 2—CycloneCC process flow diagram illustrating the complete workflow of CycloneCC

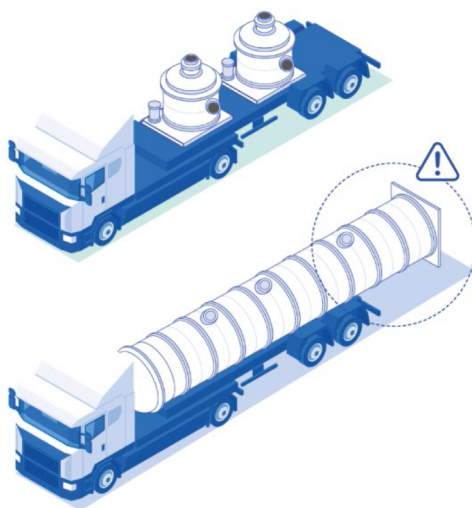


Figure 3—RPB vs conventional absorber transportation illustrating the difficulty of conventional column logistics.

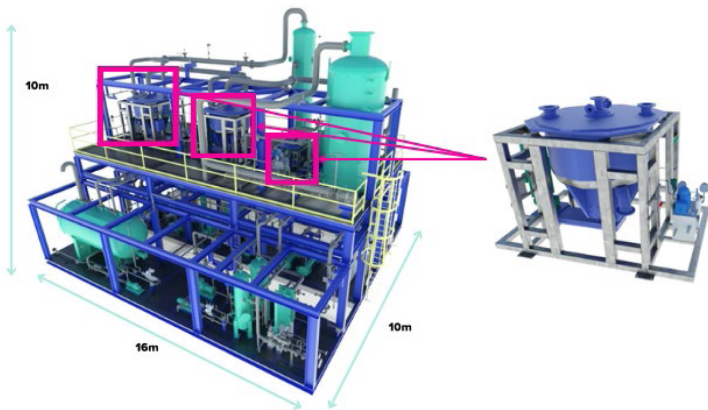


(a)



(b)

Figure 4—Lab scale RPB setup (a) front view of RPB (b) back view of RPB.



(a)



(b)

Figure 5—(a) CycloneCC industrial demonstration unit design and (b) CycloneCC industrial demonstration unit

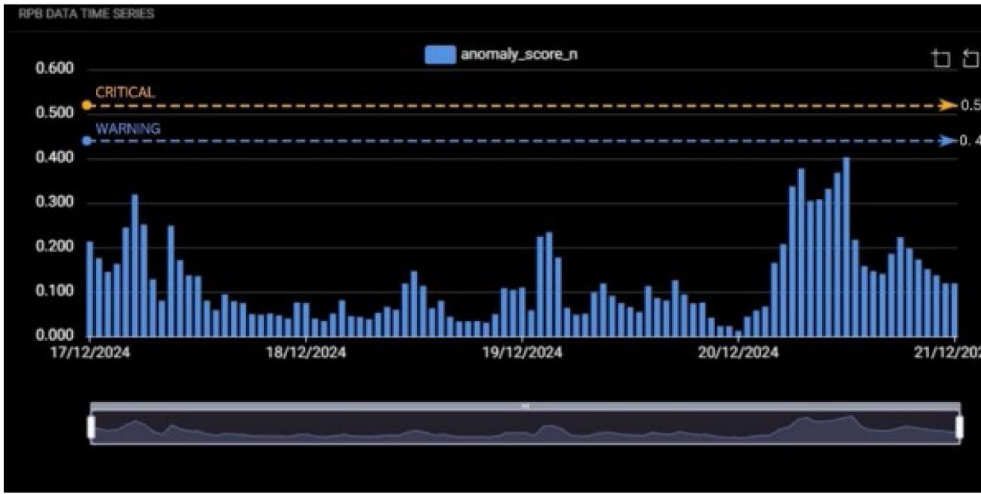


Figure 6—Example RPB anomaly dashboard illustrating the described anomaly that took place in December

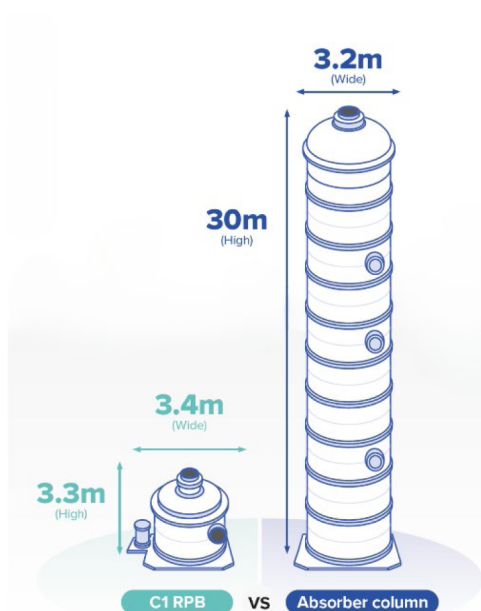


Figure 7—C1 RPB comparison to conventional absorber columns, demonstrating the size reduction



Figure 8—Mico pilot carbon capture plant within the GIC



Figure 9—C1 RPB at Thomas Broadbent, UK (picture taken during WET Factory Acceptance Testing)